Just think of the number of times you change your cellphone. Or buy yourself a new TV, computer or air-conditioner. But do you ever stop to think what becomes of the old gadgets that you are replacing? You only have to visit Mustafabad in northeast Delhi to find that out. One of India's largest electronic graveyards, shops and godowns here exist cheek by jowl in narrow, densely populated bylanes. Enter one, and you will find green, fully-wired printed circuit boards stacked to the roof and consoles of TV sets overflowing into the street. In another, air-conditioner compressors and other parts are being stripped of their metal. A third bylane is full of copper wire jutting out of white gunny sacks. An orchestra of grinding, crunching or soldering sounds plays all around you, even as the putrid smell of burning plastic hangs permanently in the air. Every part of an electronic device comes apart here, yielding not just iron, copper or aluminium but also precious gold, silver and platinum, every last bit extracted to be sold. There's a hundred times more gold in a tonne of electronic waste than in a tonne of ore—or so they say. It's a fact that a tonne of discarded cellphones or PCs can give you 280 grams of gold, worth Rs 16 lakh.

That's the loot your local kabadi centres are probably after. Electronic or e-waste has become a big business and a big threat in India in the past few decades. It covers the three broad categories of electronic goods—white (think refrigerators, washing machines and air-conditioners); grey (desktop computers, laptops, cellphones and printers); and brown

Graphics by TANMOY CHAKRABORTY

DECODING E-WASTE

ASSESSING THE NATURE OF THE BEAST THAT IS E-WASTE AND THE HARM BAD MANAGEMENT CAN INFLICT ON THE HUMAN BODY AND ENVIRONMENT

WHAT IS E-WASTE

Electronic waste or e-waste is essentially old, end-of-life electrical and electronic equipment (EEE) that users have discarded

TYPES OF E-WASTE

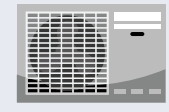
These are divided into three categories

WHITE

Refrigerators, washing machines and air-conditioners

PROJECTED MARKET SIZE
(in million units)

Air Conditioners



6.5
16.5

Refrigerators



14.5
27.5

Washing machines



7
12.6

FY19 FY25

BROWN

Televisions, cameras and recorders

PROJECTED MARKET SIZE
(in million units)

Televisions



17.5
28.4

Audio



19.5
48.5

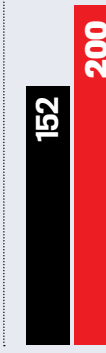
FY19 FY25

GREY

Computers, laptops, cellphones and printers

PROJECTED SHIPMENTS OF SMART-PHONES
(in million units)

2019
2025



AVERAGE PRODUCT LIFE FOR COMPUTING E-WASTE GENERATION

Televisions: **9 years**
Washing machines: **9 years**
Air-conditioners: **10 years**
Refrigerators: **10 years**
Smartphones: **5 years**

Source: CEAMA-Frost & Sullivan Report/ IDC